Movies API 1.0.0 OAS 3.0

/api/schema/

A RESTful API for managing a movie catalogue with reviews and analytics. Supports full CRUD for movies and reviews, token-based authentication, and analytical endpoints for ratings, genres, and decade trends.

MIT

Authorize

auth

POST /api/auth/login/

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema

```
{
  "username": "string",
  "password": "string"
}
```

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. **Example Value** Schema	*No links*

Code	Description	Links
	`{ "token": "string" }`	

POST /api/auth/register/

Parameters Try it out

No parameters

Request body required application/json

Example Value Schema

```
{
  "username": "string",
  "password": "string"
}
```

Responses

Code	Description	Links
201	Media type application/json Controls Accept header. **Example Value** Schema `{ "token": "string" }`	*No links*

movies

GET /api/movies/

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
201	Media type **application/json** **Example Value** Schema	*No links*


```
[
  {
    "id": 0,
    "title": "string",
    "genre": "string",
    "release_year": 9223372036854776000,
    "director": "string",
    "overview": "string",
    "average_rating": 0,
    "review_count": 0,
    "created_at": "2026-03-20T21:13:14.721Z"
  }
]
```

```
{
  "id": 0,
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string",
  "average_rating": 0,
  "review_count": 0,
  "created_at": "2026-03-20T21:13:14.721Z"
}
```


POST `/api/movies/`

Parameters

Try it out

No parameters

Request body required

application/json

Example Value Schema

```
{
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string"
}
```

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*
201	Media type **application/json** **Example Value** Schema	*No links*

```
[
  {
    "id": 0,
    "title": "string",
    "genre": "string",
    "release_year": 9223372036854776000,
    "director": "string",
    "overview": "string",
    "average_rating": 0,
    "review_count": 0,
    "created_at": "2026-03-20T21:13:14.722Z"
  }
]
```

```
{
  "id": 0,
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string",
  "average_rating": 0,
  "review_count": 0,
  "created_at": "2026-03-20T21:13:14.723Z"
}
```

GET /api/movies/{id}/

Parameters

Try it out

Name	Description
id * required integer *(path)*	id

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. Example Value Schema	*No links*


```
{
  "id": 0,
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string",
  "average_rating": 0,
  "review_count": 0,
  "created_at": "2026-03-20T21:13:14.724Z"
}
```


PUT /api/movies/{id}/

Parameters

Try it out

Name	Description
id * required integer *(path)*	id

Request body required

application/json

Example Value Schema


```
{
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string"
}
```


Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header. **Example Value** Schema	*No links*


```
{
  "id": 0,
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string",
  "average_rating": 0,
  "review_count": 0,
  "created_at": "2026-03-20T21:13:14.725Z"
}
```


PATCH `/api/movies/{id}/`

Parameters

Try it out

Name	Description
id * required `integer` *(path)*	id

Request body **application/json**

Example Value Schema

```
{
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string"
}
```

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. **Example Value** Schema	*No links*

```
{
  "id": 0,
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string",
  "average_rating": 0,
  "review_count": 0,
  "created_at": "2026-03-20T21:13:14.726Z"
}
```

DELETE /api/movies/{id}/

Parameters

Try it out

Name	Description
id * required integer *(path)*	id

Responses

Code	Description	Links
200	Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*


```
{
  "id": 0,
  "title": "string",
  "genre": "string",
  "release_year": 9223372036854776000,
  "director": "string",
  "overview": "string",
  "average_rating": 0,
  "review_count": 0,
  "created_at": "2026-03-20T21:13:14.727Z"
}
```


ai

GET `/api/movies/{id}/recommendations/`

Returns the 5 most similar movies using a weighted content-based filtering algorithm.

Scoring breakdown (max 100 points):

- Genre match : up to 40 pts (proportional to overlapping genres)
- Director match : 30 pts (exact match, ignores 'Unknown')
- Era proximity : 20 pts (within 5 years) or 10 pts (within 15 years)
- Rating quality : 10 pts (average rating within 0.5 stars)

Performance: candidates are pre-filtered by genre or director before scoring, avoiding a full table scan of 4800+ movies.

Parameters Cancel

Name	Description
id * required `integer` *(path)*	1

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'https://devanshsinghal0109.pythonanywhere.com/api/movies/1/recommendations/' \
  -H 'accept: application/json'
```

Request URL

```
https://devanshsinghal0109.pythonanywhere.com/api/movies/1/recommendations/
```

Server response

Code	Details
200	

Response body


```
{
  "movie": {
    "id": 1,
    "title": "Avatar",
    "year": 2009,
    "genre": "Action/Adventure",
    "director": "James Cameron"
  },
  "recommendations": [
    {
      "id": 588,
      "title": "The Abyss",
      "year": 1989,
      "genre": "Adventure/Action",
      "director": "James Cameron",
      "average_rating": 3.8,
      "review_count": 4,
      "match_score": 80,
      "reasons": [
        "Similar genres (Adventure, Action)",
        "Also directed by James Cameron"
      ]
    },
    {
      "id": 5
```


Download

Response headers

```
allow: OPTIONS,GET
connection: keep-alive
content-encoding: gzip
content-type: application/json
cross-origin-opener-policy: same-origin
date: Fri,20 Mar 2026 21:12:37 GMT
referrer-policy: same-origin
server: PythonAnywhere
transfer-encoding: chunked
vary: Accept-Encoding,Accept,Cookie
x-clacks-overhead: GNU Terry Pratchett
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. Example Value Schema	No links

```
{
  "movie": {
    "additionalProp1": "string",
    "additionalProp2": "string",
    "additionalProp3": "string"
  },
  "recommendations": [
    "string"
  ],
  "algorithm": "string"
}
```

reviews

GET /api/movies/{id}/reviews/

Parameters

Try it out

Name	Description
id * required integer (path)	id

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. Example Value Schema	No links

```
[
  {
    "id": 0,
    "movie": 0,
```

Code	Description	Links


```
    "reviewer_name": "string",
    "rating": 5,
    "comment": "string",
    "created_at": "2026-03-20T21:13:14.731Z"
  }
]
```


POST /api/movies/{id}/reviews/

Parameters Try it out

Name	Description
id * required integer *(path)*	id

Request body required application/json

Example Value Schema


```
{
  "movie": 0,
  "reviewer_name": "string",
  "rating": 5,
  "comment": "string"
}
```


Responses

Code	Description	Links
201	Media type application/json Controls Accept header. **Example Value** Schema	*No links*


```
[
  {
    "id": 0,
    "movie": 0,
    "reviewer_name": "string",
    "rating": 5,
    "comment": "string",
```

Code	Description	Links


```
    "created_at": "2026-03-20T21:13:14.732Z"
  }
]
```


GET /api/movies/{id}/reviews/{review_pk}/

Parameters Try it out

Name	Description
id * required integer *(path)*	id
review_pk * required integer *(path)*	review_pk

Responses

Code	Description	Links
200	Media type **application/json** Controls Accept header. **Example Value** Schema	*No links*


```
{
  "id": 0,
  "movie": 0,
  "reviewer_name": "string",
  "rating": 5,
  "comment": "string",
  "created_at": "2026-03-20T21:13:14.733Z"
}
```


PUT /api/movies/{id}/reviews/{review_pk}/

Parameters Try it out

Name	Description
id * required integer *(path)*	id
review_pk * required integer *(path)*	review_pk

Request body required

application/json

Example Value Schema

```
{
  "movie": 0,
  "reviewer_name": "string",
  "rating": 5,
  "comment": "string"
}
```

Responses

Code	Description	Links

200	Media type
	application/json
	Controls Accept header.
	
	Example Value Schema
	
	{
	"id": 0,
	"movie": 0,
	"reviewer_name": "string",
	"rating": 5,
	"comment": "string",
	"created_at": "2026-03-20T21:13:14.734Z"
	}
		No links

PATCH /api/movies/{id}/reviews/{review_pk}/

Parameters Try it out

Name	Description
id * required integer *(path)*	id
review_pk * required integer *(path)*	review_pk

Request body **application/json**

Example Value Schema

```
{
  "movie": 0,
  "reviewer_name": "string",
  "rating": 5,
  "comment": "string"
}
```

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header. **Example Value** Schema `{ "id": 0, "movie": 0, "reviewer_name": "string", "rating": 5, "comment": "string", "created_at": "2026-03-20T21:13:14.735Z" }`	*No links*

DELETE /api/movies/{id}/reviews/{review_pk}/

Parameters Try it out

Name	Description
id * required integer *(path)*	id
review_pk * required integer *(path)*	review_pk

Responses

Code	Description	Links

200	Media type
	application/json
	Controls Accept header.
	Example Value Schema
	{
	"id": 0,
	"movie": 0,
	"reviewer_name": "string",
	"rating": 5,
	"comment": "string",
	"created_at": "2026-03-20T21:13:14.736Z"
	}
		No links

analytics

GET /api/movies/genre-summary/

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'https://devanshsinghal0109.pythonanywhere.com/api/movies/genre-summary/' \
  -H 'accept: application/json'
```

Request URL

```
https://devanshsinghal0109.pythonanywhere.com/api/movies/genre-summary/
```

Server response

Code	Details
200	**Response body**


```
[
  {
    "genre": "Action",
    "movie_count": 64,
    "average_rating": 2.75
  },
  {
    "genre": "Action/Adventure",
    "movie_count": 775,
    "average_rating": 2.9858064516129033
  },
  {
    "genre": "Action/Animation",
    "movie_count": 5,
    "average_rating": 3.4
  },
  {
    "genre": "Action/Comedy",
    "movie_count": 371,
    "average_rating": 2.981132075471698
  },
  {
    "genre": "Action/Crime",
    "movie_count": 301,
    "average_rating": 3.0697674418604652
```


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Response headers

```
allow: OPTIONS,GET
connection: keep-alive
content-encoding: gzip
content-type: application/json
cross-origin-opener-policy: same-origin
date: Fri,20 Mar 2026 21:13:14 GMT
referrer-policy: same-origin
server: PythonAnywhere
transfer-encoding: chunked
vary: Accept-Encoding,Accept,Cookie
x-clacks-overhead: GNU Terry Pratchett
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	Media type	*No links*

Code	Description	Links
	application/json Controls `Accept` header. **Example Value** Schema	

```
[
  {
    "genre": "string",
    "movie_count": 0,
    "average_rating": 0
  }
]
```

GET /api/movies/most-reviewed/

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Media type application/json Controls `Accept` header. **Example Value** Schema	*No links*

```
[
  {
    "id": 0,
    "title": "string",
    "genre": "string",
    "release_year": 9223372036854776000,
    "director": "string",
    "overview": "string",
    "average_rating": 0,
    "review_count": 0,
    "created_at": "2026-03-20T21:13:14.761Z"
  }
]
```

GET /api/movies/search/

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	Media type application/json Controls Accept header. Example Value Schema	*No links*


```
[
  {
    "id": 0,
    "title": "string",
    "genre": "string",
    "release_year": 9223372036854776000,
    "director": "string",
    "overview": "string",
    "average_rating": 0,
    "review_count": 0,
    "created_at": "2026-03-20T21:13:14.762Z"
  }
]
```


GET /api/movies/top-rated/

Parameters

Cancel

No parameters

Execute Clear

Responses

Curl

```
curl -X 'GET' \
  'https://devanshsinghal0109.pythonanywhere.com/api/movies/top-rated/' \
  -H 'accept: application/json'
```

Request URL

https://devanshsinghal0109.pythonanywhere.com/api/movies/top-rated/

Server response

Code	Details
200	

Response body


```
[
  {
    "id": 464,
    "title": "Déjà Vu",
    "genre": "Romance/Drama",
    "release_year": 1998,
    "director": "Henry Jaglom",
    "overview": "L.A. shop owner Dana and Englishman Sean meet and fal
l in love at first sight, but Sean is married and Dana is to marry her
business partner Alex.",
    "average_rating": 5,
    "review_count": 1,
    "created_at": "2026-03-20T04:46:26.847087Z"
  },
  {
    "id": 2380,
    "title": "Restoration",
    "genre": "Horror",
    "release_year": 2016,
    "director": "Zack Ward",
    "overview": "During home renovations, a young couple release a fie
ry spirit seeking retribution. To save themselves and set the spirit f
ree, they must uncover the dire truth. But nothing is as s    e
seems...",
    "average_rating": 5
```


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Response headers

```
allow: OPTIONS,GET
connection: keep-alive
content-encoding: gzip
content-type: application/json
cross-origin-opener-policy: same-origin
date: Fri,20 Mar 2026 21:13:03 GMT
referrer-policy: same-origin
server: PythonAnywhere
transfer-encoding: chunked
vary: Accept-Encoding,Accept,Cookie
x-clacks-overhead: GNU Terry Pratchett
x-content-type-options: nosniff
x-frame-options: DENY
```

Responses

Code	Description	Links
200	Media type **application/json** Controls `Accept` header.	*No links*

Example Value Schema


```
[
  {
    "id": 0,
    "title": "string",
    "genre": "string",
    "release_year": 9223372036854776000,
```

Code	Description	Links


```
    "director": "string",
    "overview": "string",
    "average_rating": 0,
    "review_count": 0,
    "created_at": "2026-03-20T21:13:14.765Z"
  }
]
```


Schemas

GenreSummary {

genre*	[...]
movie_count*	[...]
average_rating*	[...]

}

LoginRequest {

username*	[...]
password*	[...]

}

Movie {

id*	[...]
title*	[...]
genre*	[...]
release_year*	[...]
director*	[...]
overview	[...]
average_rating*	[...]
review_count*	[...]
created_at*	[...]

}

PatchedMovie {

```
    id                    [...]
    title                 [...]
    genre                 [...]
    release_year          [...]
    director              [...]
    overview              [...]
    average_rating        [...]
    review_count          [...]
    created_at            [...]
}
```

PatchedReview {

```
    id                    [...]
    movie                 [...]
    reviewer_name         [...]
    rating                [...]
    comment               [...]
    created_at            [...]
}
```

RecommendationResponse {

```
    movie*                {...}
    recommendations*      [...]
    algorithm*            [...]
}
```

RegisterRequest {

```
    username*             [...]
    password*             [...]
}
```

```
Review {
    id*                    [...]
    movie*                 [...]
    reviewer_name*         [...]
    rating*                [...]
    comment                [...]
    created_at*            [...]
}

TokenResponse {
    token*                 [...]
}

TokenResponse2 {
    token*                 [...]
}
```